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Visual Object Recognition with Unknown Input Detection

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Abstract

Keywords: Out-of-Distribution Detection (OOD), Visual Object Recognition, Vision Transformers (ViTs), DINO, Post-Hoc Methods, Hierarchical Detection, Feature Representation, Unknown Input Detection

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Abstrakt

Klíčová slova: Detekci mimo distribuci (OOD), Rozpoznávání vizuálních objektů, Vizuální transformátory (ViTs), DINO, Post-hoc metody, Hierarchická detekce, Reprezentace příznaků, Detekce neznámých vstupů

Překlad názvu: Rozpoznávání vizuálních objektů s detekcí neznámých vstupů

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Chapter 1

Introduction

Deep learning architectures are currently experiencing a substantial surge in prevalence. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have demonstrated remarkable performance in various computer vision tasks. However, the real world is full of unknowns, which poses serious challenges for these models. When deployed in dynamic environments, they often encounter inputs that differ significantly from their training data, leading to unreliable predictions, **as illustrated in Figure 1.1.** The task of identifying such inputs, known as out-of-distribution (OOD) detection, is crucial for ensuring the safety and reliability of AI systems.

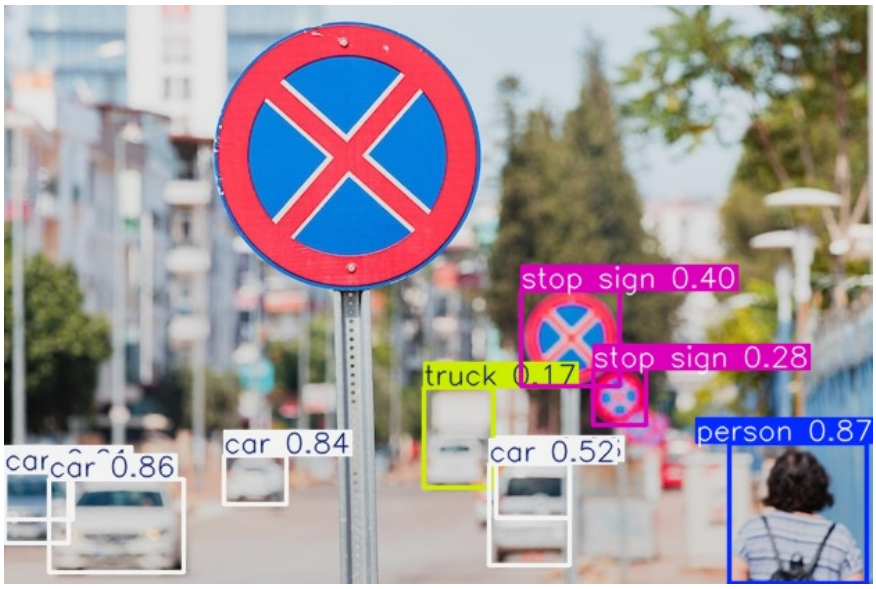


Figure 1.1: Limitations of a standard YOLO detector. The model fails to recognize the nearest 'No Parking' sign (False Negative) while misclassifying distant ones as 'Stop Signs'. This illustrates the necessity for robust feature representations.

Previous research has primarily focused on binary OOD detection, where models must distinguish only between in-distribution (ID) and OOD samples.[1] While this is an important step towards robust AI, it does not fully capture the complexity of real-world scenarios. For example, when an au-

tonomous vehicle encounters an unseen input, it is insufficient to simply flag it as OOD. Instead, a hierarchical approach assigns it to a generic class (e.g., 'obstacle'), preserving semantic information essential for decision-making.

To address these challenges, this thesis explores the capabilities of self-supervised Vision Transformers, specifically the DINOv3 architecture[2]. We selected this model for its state-of-the-art performance on out-of-distribution benchmarks (as shown in ObjectNet evaluations) and its ability to produce rich semantic representations while being kept frozen. This "frozen" capability allows for substantial computational savings, making the model ideal for efficient **post-hoc** analysis without the need for extensive fine-tuning.

1.1 Contributions

The following sections summarize the main contributions of this thesis:

1.

2.

3.

Chapter 2

Related Work

This chapter reviews the existing literature relevant to the thesis. It covers the evolution of visual backbone architectures focused on Vision Transformers and Self-Supervised Learning. In due course, it delves into established methodologies for OOD detection. Finally, it explores the emerging concept of hierarchical OOD detection, highlighting its significance and the challenges it presents.

2.1 Visual Representation Learning

State-of-the-art computer vision architectures have evolved significantly from traditonal CNNs to more sophisticated models like ViTs.

2.1.1 Vision Transformers (ViT)

Vision Transformers (ViTs)[3] have revolutionized and dethroned CNNs as the dominant architecture in computer vision. Unlike CNNs, which process images via local receptive fields and convolutional layers, ViTs leverage the self-attention mechanism from the Transformer architecture[4] to capture global dependencies in images.

2.1.2 Self-Supervised Learning and DINO

Training ViTs typically requires massive labeled datasets. To mitigate this, Self-Supervised Learning (SSL) has emerged as a powerful alternative. Methods like DINO (Distillation with No Labels)[5] and its successors, DINOv2 and DINOv3[2][6], learn robust features by enforcing consistency between different views of the same image.

2.2 Out-of-Distribution Detection

The aim of OOD detection is to safely distinguis between data that the model has seen during training (ID) and data that is novel or anomalous (OOD).

2.2.1 Post-Hoc OOD Methods

Post-hoc OOD detection methods derive anomaly scores directly from a pre-trained model without the need for parameter modification or retraining. This property makes them computationally efficient and easily deployable in real-world scenarios.

Maximum Softmax Probability (MSP): The fundamental approach proposed by Hendrycks and Gimpel[7] assumes that a classifier's softmax output can be used to gauge uncertainty. Inputs with low maximum softmax probabilities are flagged as OOD, indicating that the model is less confident in its predictions.

Distanced based Methods: To mitigate overconfidence, these methods assess the distance of an input's representation from known class distributions. Lee at al.[8] proposed using the **Mahalanobis Distance**, modeling the pre-softmax features of each class as a multivariate Gaussian distribution. An input is flagged as OOD if it lies far from the nearest class centroid in this latent space.

Energy-Based Models: Liu et al.[9] introduced an energy-based scoring function derived from the model's logits. The energy score provides a more calibrated measure of uncertainty, with lower energy values indicating higher confidence in the input being ID.

ODIN (Out-of-Distribution detector for Neural networks): Liang et al.[10] introduced ODIN, which improves upon the MSP baseline by combining two strategies: *temperature scaling* and *input perturbation*.

- First, a large temperature scalar is applied to the softmax function to separate the probability distributions of ID and OOD samples.
- Second, small perturbations are added to the input image in the direction of the gradient to aggressively maximize the predicted softmax score for ID samples.

While effective, the requirement for input preprocessing increases computational cost during inference compared to pure output-based methods.

2.2.2 Training Based OOD Methods

While post-hoc methods operate on fixed models, training-based methods explicitly optimize the model parameters to enhance the separability between ID and OOD data during the learning phase.

Ensemble Methods: Deep Ensembles[11] utilizes multiple models trained independently to capture uncertainty. The variance in predictions across the ensemble serves as an indicator of OOD inputs, with higher variance suggesting greater uncertainty. While this approach significantly improves uncertainty estimation and OOD detection accuracy, it comes with a high computational cost (both for training and inference) that is often prohibitive for real-time applications.

Regularized Training: Other methods introduce specific loss functions or *proxy* OOD data during training. For instance, CSI [12] utilizes contrastive

learning with strong data augmentations to simulate distributional shifts. Similarly, Adversarial Reciprocal Points Learning (ARPL) [13] generates *reciprocal points* in the latent space to represent the unknown class. Although ARPL achieves state-of-the-art performance on many benchmarks, it requires a complex training scheme and extensive hyper-parameter tuning [14].

Outlier Exposure (OE): A direct approach involves training the model on both ID data and a large auxiliary dataset of known outliers (e.g., 80 Million Tiny Images). The model is penalized if it predicts high confidence for these outliers [15]. While effective, this method relies on the availability of a curated OOD dataset and may not generalize well to anomalies that are distributionally distinct from the auxiliary data.

2.3 Hierarchical and Open-Set Recognition

Standard OOD treats all unknown inputs as a single homogeneous class. Nevertheless, in many real-world applications it is beneficial to understand semantic nature of unrecognized objects.

2.3.1 From Open-Set to Open-World

Open-Set Recognition (OSR) extends long-established classification by requiring models to accurately classify known classes while rejecting unknown classes during inference[16].

Open-World Recognition (OWR) inclines to more a realistic approach, where the system must identify novel inputs and eventually incorporate them into its knowledge base through continuous learning[17].

2.3.2 Hierarchical Recognition

In countless real-world scenarios, classes follow a taxonomic structure. For instance, in biological classification, species are grouped into genera, families, orders, and so on. Hierarchical recognition leverages this structure to provide more informative predictions. Instead of a flat classification where all errors are equal (e.g., confusing a "dog" with a "car" is the same as confusing a "dog" with a "wolf"), hierarchical models utilize structured label spaces. This allows the system to perform *coarse-to-fine* inference. As illustrated in Figure 2.1 using the Oxford-IIIT Pet dataset [18], classes can be naturally grouped into super-categories (e.g., Felines vs. Canines).

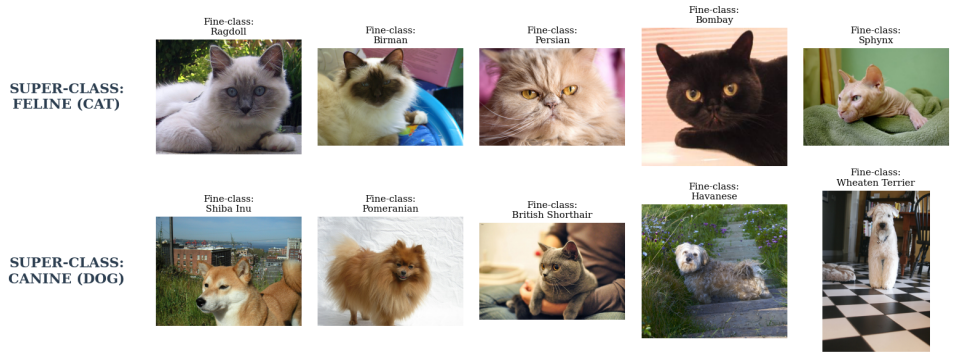


Figure 2.1: Visual demonstration of semantic hierarchy. Fine-grained categories (specific breeds) naturally cluster into broader super-classes (Felines and Canines). A hierarchical OOD detector can leverage this structure to classify an unknown dog breed as a generic 'Canine' rather than rejecting it completely.

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Appendix A

List of Abbreviations

ViT	Vision Transformer
CNN	Convolutional Neural Network
OOD	Out-of-Distribution
DINO	Self-Distillation with No Labels
ID	In-Distribution
SSL	Self-Supervised Learning
MSP	Maximum Softmax Probability
OE	Outlier Exposure
OSR	Open-Set Recognition
OWR	Open-World Recognition
ARPL	Adversarial Reciprocal Points Learning